

A CONDITIONAL SIX-QUANTIFIER DEFINITION OF $\mathbb{Z}$ INSIDE $\mathbb{Q}$: MODEL-FIELD ASSEMBLY, ITS WALLS, AND A MACHINE-VERIFIED FINITE PROGRAM

H10Q PROJECT

ABSTRACT. We present a six-quantifier architecture for a universal definition of $\mathbb{Z}$ in $\mathbb{Q}$ under an explicit member-existence hypothesis, while making no new quantifier-record claim. The class side is now unconditional: for every cell (w, z) , a character-sum choice of a , an exact Chinese-remainder system, and Dirichlet's theorem produce infinitely many aligned base classes on the square branch. For any one such class whose linear/octic pair is Schinzel-admissible, classical Schinzel Hypothesis H implies infinitely many ladder-zero members. The admissibility audit proves uniformly that the linear progression is primitive, the pair has no fixed prime divisor, the normalized octic is a unit at every frozen prime, its frozen square classes are constant, and its leading coefficient is positive. It also corrects the normalization premise: only the square class of the content, not its literal support, must be supported on the frozen set.

One algebraic condition remains: irreducibility of the degree-8 polynomial $P(a, z)$ is open uniformly in the constructed parameters. Exact modular certificates prove it for 353/353 reconstructed grid classes and 706/706 legacy grid rows, but those finite results do not prove the all-cell statement. Consequently the present chain does *not* show that Schinzel's Hypothesis H implies the six-quantifier definition; uniform irreducibility is the single named residual gap. Separately, the canonical 5-wall is always breakable on the class side, and the last searched cell $w = 179$ now has a certified member, so every off-grid cell tried to date is closed instance by instance.

CONTENTS

1. Introduction	2
1.1. The record chain and its audit	3
1.2. Contributions and scope	3
2. Model-field architecture: selection, witness tying, and the open norm problem	4
2.1. The finite-field selector is exact	4
2.2. The tied model and its six-variable bookkeeping	5
2.3. Local structure and the exact global obstruction	6
2.4. Structural corollaries and the remaining lemma	6
3. Obstruction walls and reciprocity steering	7
3.1. The alignment wall	7
3.2. Reciprocity steering	7
3.3. The generalized coprimality wall	8
3.4. Composite squarefree wall	9
4. Class Dichotomy, Characterization, and Escapes	9
4.1. The L10 dichotomy	10
4.2. Free branches, an immovable wall	10
4.3. Non-coprime escapes and complete grid coverage	11
4.4. Why no square-class shortcut remains	11
5. Uniform existence of an aligned class	11
6. The verified quantitative layer	13

6.1. L14: instance verification	14
6.2. L15: the exact remainder and counting laws	14
6.3. L16: why the small layer has sieve shape	14
6.4. L17: matched validation	15
6.5. L17 cofactor layer: reciprocity and its limit	15
7. The conditional record and the Schinzel audit	16
7.1. The statement, with its hypothesis exposed	16
7.2. What is proved, and what Schinzel must supply	16
7.3. PROVED finite audit of the Schinzel pair	17
7.4. Residual certificates and the literature boundary	17
8. Schinzel's Hypothesis H: the fixed-class implication and its gap	18
8.1. The operational contract	18
8.2. The theorem	18
8.3. Uniform admissibility status	19
9. The step-(ii) density law and the off-grid frontier	20
9.1. An unconditional local bound	20
9.2. The off-grid horizon	20
10. Conclusion and outlook	22
Reproducibility	22
References	22

1. INTRODUCTION

Hilbert's Tenth Problem over $\mathbb{Q}$ asks for an algorithm which, given a finite system of polynomials with rational coefficients, decides whether the system has a common rational zero. The problem remains open in both directions. One standard route to undecidability would be an existential definition of $\mathbb{Z}$ in $\mathbb{Q}$, since such a definition would transport the undecidability over $\mathbb{Z}$ to rational points. The converse is not known: failure of that route would not imply decidability over $\mathbb{Q}$.

There is already a sharp first obstruction. Cornelissen–Zahidi proved that $\mathbb{Z}$ cannot be the projection of the rational points of a curve over $\mathbb{Q}$; in particular, no polynomial with one existential witness defines $\mathbb{Z}$ in $\mathbb{Q}$ [4]. Thus two witnesses, equivalently a surface as witness variety, form the first open existential case. This is also exactly where the available topological and thin-set lower-bound methods stop. A different elliptic-curve route has its own cliff. If only finitely many primes are withheld from an S -integer ring, the indices n for which $x(nP)$ is integral are a finite union of congruence classes, relative to the chosen integral Weierstrass model. When nonempty this set has positive density and is archimedeanly equidistributed on a real component, rather than discrete. Consequently the prime-index mechanism used in elliptic Diophantine models degenerates at the cofinite- S boundary; this statement does not rule out unrelated predicates.

Koenigsmann changed the quantitative landscape by proving that $\mathbb{Z}$ is purely universally definable in $\mathbb{Q}$ [9]. Complementation turns this into the study of the existential rank $\text{rk}_{\mathbb{Q}}^{\exists}(\mathbb{Q} \setminus \mathbb{Z})$. Daans–Dittmann–Fehm then supplied the geometric interpretation: for a non-quantifier-free Diophantine set D over $\mathbb{Q}$,

$$\text{rk}_{\mathbb{Q}}^{\exists}(D) = \text{efd}_{\mathbb{Q}}(D) + 1,$$

and $\text{efd}_{\mathbb{Q}}(D) \leq d$ precisely when D is the image of a morphism all of whose fibres have dimension at most d [6]. The quantifier race is therefore a fibre-dimension problem. In particular, a six-universal-quantifier definition would amount to the bound $\text{efd}_{\mathbb{Q}}(\mathbb{Q} \setminus \mathbb{Z}) \leq 5$, not merely to more efficient logical notation.

1.1. The record chain and its audit. The refereed benchmark is Daans’s ten-quantifier universal definition [5]. Our source-level audit reconstructs its budget before applying any fusion: two quaternion parameters, three witnesses for the auxiliary Φ condition, one splitting witness, and two three-variable trace-norm certificates give twelve variables. Two uses of the intersection theorem of Daans–Dittmann–Fehm save one coordinate each, giving ten; pairwise reassociation cannot improve that proof shape. The same audit isolates the real places to seek an improvement: stronger simultaneous fusion, a cheaper trace-norm primitive, or a joint treatment of the nested valuation-ring conditions.

Sun subsequently announced an unrefereed seven-quantifier construction [13]. It retains Daans’s bridge but replaces the six-witness intersection block by a finite disjunction of three-witness quadrics, and then uses one further fusion. Our A2 audit checked the exact quaternion identities, the block’s soundness, the residue-field combinatorics, and explicit bridge certificates. It also traced the local–global completeness argument through parameterized Hensel lifting, weak and norm approximation, and Hasse–Minkowski. No substantive missing implication was found, but the posted preprint has stale cross-references at the decisive step, and its local–global chain was hand-traced rather than machine-verified. We therefore call seven an *unrefereed announcement*; ten remains the refereed record.

PROVED. The audit also corrects one ingredient of the unrefereed announcement [13]. Its finite-field selection lemma assumes a residue field larger than 25. Exhaustion first showed that the small fields do not fail; more conceptually, we proved the exact identity

$$N(\tau) = \frac{\#E_{\tau^{-2}}(\mathbb{F}_q)}{4}, \quad E_{\tau^{-2}} : y^2 = x(x-1)(x-\tau^{-2}).$$

Hasse’s bound makes this positive over every odd finite field admitting an admissible τ . Hence the threshold can be deleted: $\mathbb{F}_3$ is exceptional only because it has no $\tau \notin \{0, \pm 1\}$, while the preprint’s other, data-dependent exceptional places remain untouched.

1.2. Contributions and scope. The contributions of this paper are deliberately separated by status.

- (1) **CONDITIONAL — a six-quantifier architecture.** We tie a witness in Sun’s quadratic block to Daans’s congruence parameter by $a = 1 + 2s$. Soundness, denominator clearing, the exact target-place selector, and the global norm obstruction are proved; completeness is the explicitly named hypothesis H . If H holds, then $\mathbb{Q} \setminus \mathbb{Z}$ has existential rank at most six, $\mathbb{Z}$ has a six-quantifier universal definition, and the essential fibre dimension is at most five.
- (2) **PROVED — wall theorems.** We prove the two-adic source–target alignment wall, show that branch choice cannot remove the reciprocity obstruction, and prove the generalized wall: every admissible b whose odd support is coprime to the cell data fails. The only available door is controlled non-coprimality.
- (3) **PROVED — complete class characterization.** For the prime- b family, on every branch, an aligned class exists exactly when the numerator of z has a prime divisor congruent to 1 modulo 4. The necessity holds for every admissible rational a , not only when $1 + 4a^2$ is prime; the sufficiency is constructive.
- (4) **PROVED — uniform class existence.** For every cell, choose the fixed numerator prime $f = w$ and an odd a with $\left(\frac{1+4a^2}{w}\right) = -1$. A nonempty explicit CRT system and Dirichlet’s theorem then give infinitely many square-branch aligned classes. Class existence is therefore no longer part of the global hypothesis.
- (5) **PROVED — exhaustive finite verification.** On the declared grid, every one of the 353 cells has a replayable soluble witness, and every one of the 293 aligned classes has a certified

emergent-free member. These are bounded, per-row theorems under the proven-primality engine, not an extrapolation to all primes.

- (6) **EVIDENCE — a mechanistic sieve with matched validation.** Exact root lifting turns each recurring small-prime obstruction into an odd-valuation p -adic mass. On the identical 8,760-member sample, the matched marginal ratio is 0.9993, the hit-conditioned union residual is -2.9 , and a marginal-preserving permutation gives $z = -0.09$. These data validate the small-layer mechanism in the measured window; they do not supply a tail bound or a uniform density theorem.
- (7) **PROVED implication / OPEN uniform premise — the Schinzel reduction.** For a fixed fully admissible class, Schinzel's Hypothesis H for the linear/octic pair implies infinitely many emergent-free members. The uniform audit proves coprimality, absence of a fixed prime divisor, frozen-unit and symbol constancy, and positive leading coefficient. It does not prove irreducibility of the octic uniformly. Exact modular certificates cover 353/353 reconstructed grid classes and 706/706 legacy rows, but those finite certificates do not close the all-cell gap.
- (8) **PROVED — off-grid horizon extension.** The same certificate machinery now closes and replays $[179, [-1, 1]]$, the last searched open cell. Every off-grid cell tried to date has a certified member, but this remains a finite instance statement, not a uniform member theorem.

What is not claimed. We claim *no new unconditional quantifier record*: Daans's ten remains the refereed record, and Sun's seven remains an unrefereed announcement. Nor do we claim that classical Schinzel H presently implies the six-quantifier statement. First one must prove that the constructed degree-8 polynomial $P(a, z)$ is irreducible for every cell, or prove that the parameter choice can always avoid its reducible locus. Only after that uniform algebraic premise is supplied does Schinzel address the simultaneous prime-value member problem.

2. MODEL-FIELD ARCHITECTURE: SELECTION, WITNESS TYING, AND THE OPEN NORM PROBLEM

2.1. The finite-field selector is exact.

Theorem 2.1 (PROVED: L4, the small-field exhaustion). *Let q be an odd prime power with $q \leq 25$, let χ be the quadratic character of $\mathbb{F}_q$ with $\chi(0) = 0$, and let $\tau \in \mathbb{F}_q \setminus \{0, \pm 1\}$. There is an $a \in \mathbb{F}_q$ such that*

$$\chi(1 + 4a^2) = -1, \quad \chi(-(1 + 4a^2)(1 - \tau^2(1 + 4a^2))) = 1.$$

For $q = 3$ the assertion is vacuous, because there is no admissible τ . For $q \in \{5, 7, 9, 11, 13\}$ the minimum number of choices is 2, and for $q \in \{17, 19, 23, 25\}$ it is 4.

Thus the constant 25 in the unrefereed character-sum proof of [13] is not a genuine small-field obstruction: among odd residue fields, only the field of order 3 remains, and there the issue is admissibility rather than failure of the selector.

Theorem 2.2 (PROVED: L5, the Legendre identity). *For every odd prime power q , every $\tau \in \mathbb{F}_q \setminus \{0, \pm 1\}$, and $\lambda = \tau^{-2}$, let $E_\lambda : y^2 = x(x - 1)(x - \lambda)$. Then*

$$N(\tau) := \#\{a \in \mathbb{F}_q : \chi(1 + 4a^2) = -1, \chi(-(1 + 4a^2)(1 - \tau^2(1 + 4a^2))) = 1\} = \frac{\#E_\lambda(\mathbb{F}_q)}{4}.$$

Consequently

$$\#E_\lambda(\mathbb{F}_q) \geq (\sqrt{q} - 1)^2 > 0, \quad N(\tau) \geq (\sqrt{q} - 1)^2/4 > 0,$$

so the hypothesis $|k| > 25$ can be deleted from the lemma.

The proof is an exact point count, not an asymptotic substitution. With $u_a = 1 + 4a^2$ and $w_a = -u_a(1 - \tau^2 u_a)$, expand the corrected indicator; the hyperbola identity $\sum_b \chi(b^2 + c) = -1$ for $c \neq 0$ evaluates the quadratic pieces. Fibre-counting the map $a \mapsto u_a$ then gives

$$4N = q + 1 + \sum_u \chi(u(u-1)(u-\lambda)) = \#E_\lambda(\mathbb{F}_q).$$

This also explains divisibility by 4: the Legendre curve has its full rational 2-torsion.

2.2. The tied model and its six-variable bookkeeping. Set $K = \mathbb{Q}$, $S = \{2\}$, $\pi = 2$, and $u = 1$. Pull the bridge back along $z \mapsto z^3$ and put $A = 1 + 4a^2$, $B = 2b$, $\delta_\tau = 1 - A\tau^2$, and $c = h(a, b, z^3) = a^2 z^6 g(a, b)/(1 - z^3 - a^2 z^6)$. The original block is

$$\Psi_\tau(a, b, c) : \exists y, r, s [\delta_\tau(c^2 - Ay^2) - 16B(r^2 - As^2) = 16].$$

The architectural step is to identify its witness s with the congruence parameter by imposing $a = 1 + 2s$. We use the finite branch menu

$$\tau \in \left\{ 0, \frac{2a}{A}, \tau^\dagger = \frac{1 + 2a^2}{A} \right\}.$$

After clearing the nonzero denominators, the resulting two-witness block is

$$\begin{aligned} \Theta_*(a, b, c, s) : \quad & \exists y, r \quad [c^2 - Ay^2 - 16Br^2 = 16 - 16ABs^2] \\ & \vee [c^2 - Ay^2 - 16ABr^2 = 16A - 16A^2Bs^2] \\ & \vee [a^4(c^2 - Ay^2) + 4ABr^2 = 4A^2Bs^2 - 4A]. \end{aligned}$$

The three equations correspond respectively to $\tau = 0$, $\tau = 2a/A$, and $\tau^\dagger$; on the last branch $\delta_{\tau^\dagger} = -4a^4/A$ and $\alpha = (2a^2)^2$ is a square. A finite disjunction is represented by the product of these polynomials in the same witnesses (y, r) , so adding the square branch costs no variable. The candidate model is therefore

$$F(z) := \exists b, s [(1 + 2s, b) \in \Phi_1^{\{2\}} \wedge \Theta_*(1 + 2s, b, h(1 + 2s, b, z^3), s)].$$

Here $\Phi_1^{\{2\}}$ enforces $v_2(s) \geq 0$ and $b \in \mathcal{O}_2^\times$. Membership in it has existential rank at most 3; the tied block has rank 2 in (y, r) . The intersection theorem of [6] replaces their conjunction by $3 + 2 - 1 = 4$ witnesses, and projection of the two base coordinates (b, s) gives

$$2 + (3 + 2 - 1) = 6.$$

If an input set has rank zero, the elementary conjunction bound is no larger. The two branches share (y, r) and are combined by multiplying their equations; denominator clearing introduces no inverse witness because $A \equiv 5 \pmod{8}$ at 2 and $1 - z^3 - a^2 z^6$ cannot vanish over $\mathbb{Q}$.

For use by the later local model, write $Z = z^3$, $D_z = 1 - z^3 - a^2 z^6$, and $N_g(b) = 16a^4 b^2 - A(b-1)^4$. The denominator-cleared value polynomial is

$$P(b) = 16D_z^2 A^2 b^4 - \delta_\tau a^4 Z^4 N_g(b)^2 - 32A^3 s^2 D_z^2 b^5.$$

Finally define the tied target value

$$M_\tau = 16 - \delta_\tau c^2 - 16ABs^2.$$

PROVED: soundness. CONDITIONAL: completeness. Soundness, $F(z) \Rightarrow z \in \bigcup_{w \notin S} \mathbf{m}_w$, is proved; the converse is the open assembly lemma. If that lemma holds, the union outside S is $\exists_6$; adjoining the finitely many omitted maximal ideals preserves that count, so $\mathbb{Q} \setminus \mathbb{Z}$ is $\exists_6$, $\mathbb{Z}$ is $\forall_6$, and $\text{efd}_{\mathbb{Q}}(\mathbb{Q} \setminus \mathbb{Z}) \leq 5$.

2.3. Local structure and the exact global obstruction. PROVED (W0).

The first criterion, W0, is uniform even in the residue field $\mathbb{F}_3$. For every finite field k of odd order,

$$\sum_{a \in k} \chi(1 + 4a^2) = -1.$$

Hence some a makes A a nonsquare, and the canonical selector is

$$\tau_*(a) = \begin{cases} 0, & \chi(-1) = -1, \\ 2a/A, & \chi(-1) = 1. \end{cases}$$

It satisfies $\delta_{\tau_*} \neq 0$ and $\chi(-\delta_{\tau_*} A) = 1$: the two values of $-\delta_{\tau_*} A$ are $-A$ and -1 . Since $s \mapsto 1 + 2s$ is a bijection, tying loses no residue class and removes L5's $\mathbb{F}_3$ vacancy.

PROVED (W1).

For W1, let w be odd with $v_w(A) = v_w(\delta_\tau) = 0$, $v_w(b) = 1$, $v_w(z) \geq 1$, and $v_w(s) \geq 0$. The cube pullback gives either $c = 0$ or

$$v_w(c) = 2v_w(a) + 6v_w(z) - 2 \geq 6v_w(z) - 2 \geq 4.$$

Then the tied conic

$$-\delta_\tau A y^2 - 16B r^2 = 16 - \delta_\tau c^2 - 16ABs^2$$

is soluble over $\mathbb{Q}_w$ if and only if $\chi_w(-\delta_\tau \tilde{A}) = 1$. The right side is a unit congruent to 16; its two subtracted terms have valuations at least 8 and 1. Thus the target-place test is independent of the tied value of s .

PROVED (W2).

W2 identifies the remaining obstruction. Put $E_\tau = \mathbb{Q}(\sqrt{-\delta_\tau AB})$. The conic is equivalent to

$$N_{E_\tau/\mathbb{Q}}(\delta_\tau A y + 4r\sqrt{-\delta_\tau AB}) = -\delta_\tau A M_\tau, \quad M_\tau = 16 - \delta_\tau c^2 - 16ABs^2.$$

Its global solubility is exactly

$$M_\tau = 0 \quad \text{or} \quad -\delta_\tau A M_\tau \in N_{E_\tau/\mathbb{Q}}(E_\tau^\times).$$

The zero alternative is genuine. In the two branches the data are

τ	E_τ	M_τ
0	$\mathbb{Q}(\sqrt{-AB})$	$16 - c^2 - 16ABs^2$
$2a/A$	$\mathbb{Q}(\sqrt{-B})$	$16 - c^2/A - 16ABs^2$

2.4. Structural corollaries and the remaining lemma. PROVED (L7a). At an odd place with $v(\delta_\tau A) = v(B) = 0$ and $M_\tau \neq 0$, the tied conic is soluble when $v(M_\tau)$ is even; when that valuation is odd, it is soluble exactly when $\chi_v(-\delta_\tau AB) = 1$. If also $v(A) = 0$, the corresponding untied ternary form is universal over $\mathbb{Q}_v$. Accordingly the full candidate obstruction set is

$$\{2, \infty\} \cup \{v : v(M_\tau) \text{ odd}\} \cup \{v : v(A) \neq 0 \text{ or } v(B) \neq 0\}.$$

Coefficient-bad places are genuine and cannot be replaced merely by the support of AB , because valuations can cancel in that product.

CONDITIONAL (L7b, conditional on L6 soundness). No fixed admissible (s, b) and branch admits rational functions $Y(z), R(z) \in \mathbb{Q}(z)$ satisfying

$$-\delta_\tau A Y(z)^2 - 16B R(z)^2 = M_\tau(z)$$

identically. This rules out a uniform rational section, not pointwise coverage of all relevant rational fibres by some fixed pair.

PROVED (L7c). For $M_\tau \neq 0$, local solubility is equivalent to

$$t_v := (-16\delta_\tau AB, -\delta_\tau A M_\tau)_v = 1, \quad \prod_v t_v = 1.$$

Thus a failing tied conic is obstructed at an even number of places, at least two, and repairs must change places in pairs.

CONDITIONAL / OPEN (assembly). For every odd target w and every z with $v_w(z) \geq 1$, one must choose (s, b) with $(1 + 2s, b) \in \Phi_1^{\{2\}}$, $s \equiv (\bar{a}_w - 1)/2 \pmod{\mathfrak{m}_w}$, $v_w(b) = 1$, and one of the two branches such that $M_\tau = 0$ or $-\delta_\tau AM_\tau \in N_{E_\tau/\mathbb{Q}}(E_\tau^\times)$. Equivalently, the free b -direction must steer the quadratic field $E_\tau = \mathbb{Q}(\sqrt{-\delta_\tau AB})$ simultaneously against the moving odd-valuation places of M_τ and the moving coefficient-bad places. That weak/norm-approximation statement, not W0, W1, W2, or the quantifier bookkeeping, is the open problem.

EVIDENCE. Bounded witness-switching searches support this steering mechanism, but are not a proof of assembly and are not used to upgrade the conditional six-quantifier conclusion.

3. OBSTRUCTION WALLS AND RECIPROCITY STEERING

3.1. The alignment wall. Put

$$a = 1 + 2s, \quad A = 1 + 4a^2, \quad B = 2b, \quad P = 1 - ABs^2.$$

For the two original tied branches, the source square classes are $D_0 = P$ and $D_1 = AP$, whereas the target norm fields are $E_0 = \mathbb{Q}(\sqrt{-2Ab})$ and $E_1 = \mathbb{Q}(\sqrt{-2b})$. The exact value identities show that the branch value u_τ is a norm from $\mathbb{Q}(\sqrt{D_\tau})$, while solubility asks that Au_τ be a norm from E_τ . In both branches the mismatch is measured by

$$D_\tau \text{disc}(E_\tau) \equiv -2AbP =: \kappa \pmod{\mathbb{Q}^{\times 2}}.$$

Proposition 3.1 (L8: absorption and the parity wall). **PROVED.** *At an odd place v , assume the branch value M_τ is nonzero and each of $v(\delta_\tau A)$, $v(B)$, and $v(M_\tau)$ is even. Then the tied conic has a $\mathbb{Q}_v$ -point. Consequently an odd obstruction can occur only where A , B , or M_τ has odd valuation. On the other hand, for every admissible pair, $v_2(P) = 0$ and $v_2(\kappa) = 1$.*

Proof. After scaling the two conic variables and dividing by the even power of v contributed by M_τ , one obtains a nondegenerate binary quadratic form with unit coefficients representing a unit. Over the residue field it represents every nonzero class: in the anisotropic case it is the norm form of the quadratic extension, whose norm is surjective. Hensel lifting gives the local point. This is the absorption mechanism.

Admissibility gives $v_2(ABs^2) \geq 1$, hence P is a 2-adic unit. Thus $-2AbP$ has odd 2-adic valuation. The target discriminants $-2b$ and $-2Ab$ likewise have valuation one, so the target is always a field and can never equal the source quadratic field. $\square$

The two exact-matching devices are therefore empty on the admissible locus. Indeed, matching source and target would require $\text{disc}(E_\tau) \equiv D_\tau$, equivalently $\kappa \equiv 1$ modulo squares; splitting the target would require $\text{disc}(E_\tau)$ itself to be a square. The first condition contradicts $v_2(\kappa) = 1$, and the second contradicts the odd valuation of the target discriminant. This rules out precisely these two identities, not every possible fixed square-class relation: for a general class j , one would still have to prove that an odd-multiplicity value prime inert in $\mathbb{Q}(\sqrt{j})$ must occur, and no such theorem is asserted here.

EVIDENCE. The frozen rescues do not exhibit a further support-controlled identity: all ten have odd-multiplicity wild primes, and all measured symbols at those primes are $+1$. In the recorded session sweep, thirteen of 161 admissible pairs succeeded, while every observed obstruction set had even cardinality. These observations are alignment statistics only; they do not prove a uniform mechanism or an assembly theorem.

3.2. Reciprocity steering.

Lemma 3.2 (L9 steering lemma). **PROVED.** For $x, d \in \mathbb{Q}^\times$, let

$$T = \{2, \infty\} \cup \{q \text{ odd} : v_q(x) \neq 0 \text{ or } v_q(d) \neq 0\}.$$

Then $\prod_{v \in T} (x, d)_v = 1$. Hence, if all symbols in T except possibly one are $+1$, the remaining symbol is also $+1$.

Proof. Hilbert reciprocity gives the product over all places. Outside T , both entries are odd-adic units and their Hilbert symbol is $+1$, so only the displayed finite set remains. $\square$

For the tied conic take $x = \alpha M_\tau$, $d = \alpha B$, and $\alpha = -\delta_\tau A$. Once every controlled symbol is $+1$, a single wild odd-multiplicity prime q_0 with $v_{q_0}(d) = 0$ is not an obstruction: its symbol is forced by the lemma. Thus the useful arithmetic target is not complete smoothness, but a controlled part together with one proven prime-power cofactor.

Proposition 3.3 (L9a: the prime- b symbol). **PROVED.** Fix an admissible a , a cell (w, z) , and a branch τ . Let $b = \varepsilon q_1$, where q_1 is odd prime, and impose separately

$$v_{q_1}(\delta_\tau) = v_{q_1}(A) = v_{q_1}(a) = v_{q_1}(z) = v_{q_1}(D_z) = 0.$$

Then

$$(x, d)_{q_1} = \left(\frac{A}{q_1} \right).$$

In particular, the moving place is aligned by choosing a residue class with $\left(\frac{A}{q_1} \right) = +1$.

Proof. Clearing denominators and reducing at $b \equiv 0 \pmod{q_1}$ gives $N_g(0) = -A$. The hypotheses make this a unit, the denominator contributes valuation four, and $v_{q_1}(x) = -4$. Its unit part is A modulo squares, while $v_{q_1}(d) = 1$; the Hilbert-symbol parity formula gives the stated Legendre symbol. Quadratic reciprocity makes it constant on suitable congruence classes. $\square$

The corrected progression freezes only genuine controlled places. With $S_0 = \{2\} \cup \text{supp}(\alpha)$, the modulus contains the required 2-adic and A -character moduli and the Taylor exponents at places of S_0 ; it does not freeze the moving prime. For a coprime prime q_1 in the selected class,

$$\text{supp}(d) = S_0 \cup \{q_1\}, \quad \left(\frac{A}{q_1} \right) = +1.$$

Thus every place in the support of d is controlled, and only the odd-multiplicity primes of the remaining cofactor of x are wild.

The need for the corrected modulus is witnessed by the recorded cell $(w, u) = (73, 5)$. With $s = 0$, $b_0 = -29$, $\tau = 2/5$, the superseded recipe gives $N = 689120$. The prime $q = 244637629 = 29 + 355N$ has the same prescribed class, sign, and value of $(A \mid q)$, but at 59 the symbol changes: $v_{59}(x(b_0)) = 0$ with symbol $+1$, whereas $v_{59}(x(-q)) = 1$ with symbol -1 .

CONDITIONAL. Turning such a progression into an infinite assembly family still requires the named Schinzel–H/Bunyakovsky input that simultaneously makes q_1 prime and leaves a single prime cofactor of x . Reciprocity removes the last symbol once that input is supplied; it does not prove the prime-value assertion.

3.3. The generalized coprimality wall.

Theorem 3.4 (L12a). **PROVED.** Let a be admissible and suppose $A = 1 + 4a^2$ is prime. Let τ be any branch, assume $v_A(z) \leq 0$, and suppose every odd-valuation place of an admissible rational b is coprime to $\{2, z, D_z, a, \delta_\tau, A\}$. If b_{sf} denotes the squarefree part of its odd part, then

$$\prod_{p \in \{A\} \cup \text{oddsupp}(b)} (x, d)_p = -1.$$

Thus at least one of these places obstructs every branch.

Proof sketch. At each odd place p of b , whether p occurs in the numerator or denominator, the dominant-term calculation makes $v_p(x)$ even and its unit part

$$u_x \equiv A \pmod{\mathbb{Q}_p^{\times 2}}.$$

Therefore $(x, d)_p = (A | p) = (p | A)$, and multiplication over these places gives $(b_{\text{sf}} | A)$. At A , the branch-uniform parity calculation gives $(x, d)_A = -(b_{\text{sf}} | A)$: the minus sign is $(2 | A) = -1$, since $A \equiv 5 \pmod{8}$, while the branch-dependent unit of α is a residue. The two Jacobi factors cancel and leave -1 . $\square$

The only door in this argument is non-coprimality. If b shares a prime f with the cell data, the reduction to the constant term at f and hence $u_x \equiv A$ can fail; the telescoping proof then no longer applies. This is a door, not by itself a solubility claim.

3.4. Composite squarefree wall.

Theorem 3.5 (L13e). **PROVED.** *Let a be admissible, put $A = 1 + 4a^2$, and let K be the squarefree kernel of its rational square class; this includes composite, non-squarefree, and rational A . Assume $v_p(z) \leq 0$ for every odd $p | K$ and that every odd-valuation place of an admissible rational b is coprime to $\{2, z, D_z, a, \delta_\tau, A\}$. Then, for every odd $p | K$ and every odd ℓ occurring to odd order in b ,*

$$(x, d)_p = \left(\frac{2b}{p}\right), \quad (x, d)_\ell = \left(\frac{A}{\ell}\right),$$

and

$$\prod_{p|K} (x, d)_p \prod_{\ell \in \text{oddsupp}(b)} (x, d)_\ell = \left(\frac{2}{K}\right)^{1+v_2(b)}.$$

Proof sketch. At an odd $p | K$, the middle term $-\delta_\tau a^4 Z^4 N_g^2$ strictly dominates the other terms of the value polynomial, and $v_p(N_g) = 0$. Hence $v_p(x)$ is odd and its unit class contains the complementary factor K/p . Every prime divisor of K is $1 \pmod{4}$, because it divides a value $n^2 + 4m^2$ with the data coprime. In the Hilbert-symbol formula, the complementary factors from the two entries cancel. This yields $(2b | p)$ independently of the branch.

Multiplying over $p | K$ and over the moving primes of b produces the cross terms $(M | K)(K | M)$. Quadratic reciprocity cancels them factor by factor because every divisor of K is $1 \pmod{4}$. Moreover $K \equiv A \equiv 5 \pmod{8}$, so $(2 | K) = -1$. Admissibility is exactly $v_2(b) = 0$, for which the product is -1 ; at the first non-admissible boundary $v_2(b) = 1$, the product is $+1$ and this wall dissolves. Even-exponent primes of A do not enter K . $\square$

It follows that the necessity direction of L11h is a theorem for every admissible a : escaping the coprime wall requires a prime $p | K$ with $v_p(z) > 0$, and every such p is congruent to 1 modulo 4 and divides the numerator of z . Hence reachability implies that the numerator of z has a prime divisor congruent to 1 modulo 4. This conclusion does not supply the separate prime-value input needed for assembly.

4. CLASS DICHOTOMY, CHARACTERIZATION, AND ESCAPES

Fix an admissible parameter a , put

$$A = 1 + 4a^2, \quad \delta_\tau = 1 - A\tau^2, \quad \alpha = -\delta_\tau A,$$

and first consider the prime-member family $b = \varepsilon q$, with $\varepsilon \in \{\pm 1\}$ and q prime. An *aligned class* means a nonempty prime progression in which the Hilbert symbols at every controlled finite place, at the moving place q , and at the real place are all $+1$. This is deliberately a frozen-side notion: odd-valuation primes emerging from the value of $P(b)$ are a separate obligation.

4.1. The L10 dichotomy.

Proposition 4.1 (Canonical alignment and the opposite wall; PROVED in the stated scope). *On the canonical branch $\tau = 2a/A$, the prime- b family is aligned with the canonical square class -1 in the exact sense that*

$$\delta_\tau = \frac{1}{A}, \quad \alpha = -1.$$

On the branch $\tau = 0$, if a is odd, A is prime, $A \nmid 2zD_z$, and the prime $q \neq A$ satisfies the prescribed unit conditions, then no aligned class exists.

The first assertion is an identity, not permission to discard the places above A : although α is an A -unit, $\delta_\tau = 1/A$ forces those places back into the controlled set. This distinction is essential; the obsolete grid count obtained by omitting them is not used here. For the second assertion, $\tau = 0$ gives $\delta_\tau = 1$ and $\alpha = -A$. The local calculation at A yields

$$(x, d)_A \left(\frac{A}{q} \right) = \left(\frac{-2\varepsilon}{A} \right) = -1.$$

Indeed admissibility gives $A \equiv 5 \pmod{8}$, so $\left(\frac{-1}{A} \right) = +1$ and $\left(\frac{2}{A} \right) = -1$, independently of the sign ε . Two required symbols are therefore permanently anti-correlated. This is a theorem for prime integral A under the displayed hypotheses; it is not a claim extracted from a finite failure search.

4.2. Free branches, an immovable wall. Branch completion itself costs nothing. If $\tau = p/q$ is rational in the existing parameters and its denominator is proved nonzero on the base locus, adjoining that branch introduces no inverse witness. Admissibility makes A a nonsquare in $\mathbb{Q}_2$, hence $\delta_\tau \neq 0$; the norm identity in the unrefered argument of Sun [13] is uniform in τ ; and a finite disjunction of branch equations is represented by their product in the same witnesses. Explicit choices meet all four square classes of $\langle -1, A \rangle$, including the square branch

$$\tau^\dagger = \frac{1 + 2a^2}{1 + 4a^2}, \quad \alpha = (2a^2)^2.$$

The corrected controlled set for a prime member is

$$S = \{2, 3, 5, 7\} \cup \text{supp}(\alpha) \cup \text{supp}(\delta_\tau).$$

Outside S , the fixed content of P has even valuation; after that content is removed, the reduction is a nonzero polynomial whose degree prevents a permanent odd valuation at the remaining primes. Thus no fixed bad place is silently omitted.

More importantly, changing branches does not change the obstruction. For prime A , admissible a , and $v_A(z) \leq 0$, every branch in the relevant square-class group satisfies

$$(x, d)_A (x, d)_q = -1.$$

Hence at least one controlled symbol is negative. This branch-independent wall was checked exactly on 2,136 extended instances across all four explicit branches; the computation confirms the proved identity rather than replacing it.

Theorem 4.2 (The L11h characterization; PROVED). *For the prime- b family, an aligned class exists at the cell (w, u) , with $z = wu$, if and only if the numerator of z has a prime divisor $p \equiv 1 \pmod{4}$. On the finite grid this selects 190 of 353 cells; the other 163 cells are unreachable by this family on every branch.*

Proof sketch. For necessity, the generalized wall says that alignment is impossible unless some odd prime $p \mid A$ also has $v_p(z) > 0$. Write $a = m/n$ in lowest terms. Such a prime divides $n^2 + 4m^2$, and therefore $(2mn^{-1})^2 \equiv -1 \pmod{p}$; consequently $p \equiv 1 \pmod{4}$. The factor-by-factor telescoping argument applies also when A is composite or rational, so this necessity is PROVED for every admissible a , not merely supported by a search.

Conversely, let $p \equiv 1 \pmod{4}$ divide the numerator of z . Choose odd representatives m, n satisfying $n^2 \equiv -4m^2 \pmod{p}$ and set $a = m/n$. Then a is admissible and $v_p(A) > 0$, so the factor shared by A and z breaks the wall. Freezing the complete controlled set gives an explicit aligned progression. The construction, including its escape prime, chosen a , branch, and modulus, is replayed on all 190 predicted rows. This constructive sufficiency and the preceding necessity make the equivalence the structural centre of the class analysis. $\square$

4.3. Non-coprime escapes and complete grid coverage. The characterization concerns $b = \varepsilon q$. The only effective door through the generalized coprime wall is to share a cell prime deliberately:

$$b = \varepsilon f q, \quad f \mid z.$$

This produces aligned escape classes at 103 of the 163 prime-family-walled cells. Their frozen-side alignment is PROVED row by row; the remaining 60 walled cells with no such escape are closed instead by direct, fully soluble witnesses.

The Schinzel audit also keeps six legacy residual rational/composite- b witnesses separate as pointwise certificates. They are unconditional local witnesses, but they are not promoted into prime progressions and give no prime-value theorem. The eventual last cell, $(89, -2)$, was closed by the replayed square-branch witness $(a, b, \tau) = (3, 89/367, 19/37)$, completing cell coverage.

verified object	coverage	status
L11 prime- b aligned classes	190/190	PROVED PER ROW
non-coprime escape classes	103/103	PROVED PER ROW
all aligned classes, with a zero-bad member	293/293	PROVED PER ROW
all grid cells, with a soluble witness	353/353	PROVED PER ROW

The class total is exactly the disjoint sum of the L11 and escape families; the cell total additionally uses direct pointwise witnesses. Neither total asserts a uniform theorem beyond the verified grid.

4.4. Why no square-class shortcut remains.

Proposition 4.3 (L13c; PROVED per row). *For every one of the 706 cell-branch rows, P is irreducible of degree 8 over $\mathbb{Q}$. Consequently no constant-square or weak square-factor form collapses the emergent-place obligation on the grid.*

The boundary between theorem and experiment is sharp. The irreducibility certificates are PROVED. By contrast, the 137.5 million exact square-class tests with no admissible hit are EVIDENCE against a hidden special-value shortcut, not a universal nonexistence theorem. Likewise the observed Galois group $D_8 \wr C_2$ is EVIDENCE: its cycle types matched all 28,240 certified unramified Frobenius samples, but no group-identification proof is claimed.

Finally, the earlier zero-bad search already supplied 24 certified members across 15 distinct cells; those are PROVED INSTANCES, while the rough quarter-density inferred from the sample is only EVIDENCE. The later per-class replay strengthens the finite conclusion to the 293/293 line in the table, without changing the open uniform problem.

5. UNIFORM EXISTENCE OF AN ALIGNED CLASS

The prime- b characterization of Section 4 describes one family, not the availability of classes altogether. On the square branch, controlled non-coprimality gives a class for every cell. This removes the class-existence clause from the global hypothesis; it does not produce an emergent-free member.

Theorem 5.1 (Uniform class existence; PROVED). *Let w be an odd prime and let $z \in \mathbb{Q}^\times$ satisfy $v_w(z) \geq 1$. There are an odd integer a , a fixed factor $f = w$, and infinitely many positive primes*

q_1 for which the square branch

$$A = 1 + 4a^2, \quad \tau^\dagger = \frac{1 + 2a^2}{A}, \quad \delta = -\frac{4a^4}{A}, \quad \alpha = (2a^2)^2,$$

with $\varepsilon = +1$ and $b_0 = fq_1$, is an aligned base class for the cell (w, z) . If

$$S = \{2, 3, 5, 7\} \cup \text{supp}(\alpha) \cup \text{supp}(\delta) \cup \{f\},$$

then the class freezes every symbol at S , the moving-prime symbol, and the real symbol at $+1$.

Proof. We give the construction, including the residue count. First choose a . For the quadratic character χ_f the standard character sum is

$$\sum_{r \bmod f} \chi_f(1 + 4r^2) = -1.$$

There are two zero terms if $f \equiv 1 \pmod{4}$ and none if $f \equiv 3 \pmod{4}$. Hence the number of residues with $\chi_f(1 + 4r^2) = -1$ is respectively

$$\frac{f-1}{2} \quad \text{or} \quad \frac{f+1}{2},$$

and is always positive. Choose such a residue and lift it to an odd integer a ; adding the odd modulus f changes the parity of a lift. Thus

$$(E1) \quad \left(\frac{A}{f}\right) = -1.$$

Taking $f = w$ is always admissible. Indeed, (E1) implies $f \nmid aA$, so $v_f(\delta) = 0$. With $Z = z^3$ and $D = 1 - Z - a^2Z^2$, the cell condition gives $v_f(Z) \geq 3$ and $D \equiv 1 \pmod{f}$. Thus f is an odd numerator prime, is coprime to $aA\delta D$, and meets every fixed-factor guard. This includes $f = 5$; there is no exceptional denominator or gcd case.

Put

$$M = 4A \prod_{p \in S} p$$

and require the following reduced residue system for q_1 :

$$(E2) \quad q_1 \in (\mathbb{Z}/M\mathbb{Z})^\times, \quad \left(\frac{2fq_1}{p}\right) = +1 \quad (p \in S \text{ odd}, p \neq f).$$

At each distinct odd prime in (E2), precisely half of the unit classes satisfy the displayed character. The Chinese remainder theorem therefore gives exactly

$$(E3) \quad \frac{\varphi(M)}{2^{\#\{p \in S: p \text{ odd}, p \neq f\}}}$$

reduced classes modulo M . In particular (E2) is nonempty. Dirichlet's theorem gives infinitely many positive primes q_1 in any selected class. Removing the finitely many primes dividing the fixed cell data or producing a root or degeneracy leaves infinitely many choices.

It remains to verify the local symbols. Write $s = (a-1)/2$ and

$$N_g(b) = 16a^4b^2 - A(b-1)^4,$$

so the kernel polynomial is

$$(E4) \quad P(b) = 16D^2A^2b^4 - \delta a^4Z^4N_g(b)^2 - 32A^3s^2D^2b^5.$$

Since α, D^2, A^2 , and b^4 are squares, the square class of the local conic input x is that of $P(b)$.

At f , one has $v_f(b_0) = 1$, $N_g(b_0) \equiv -A \pmod{f}$, and the three terms of (E4) have valuations 4, at least 12, and at least 5. The uniquely minimal term is a square, so x is a square in $\mathbb{Q}_f$ and the f -symbol is $+1$.

At 2 the exact computation is needed because the least unit need not be a square. The kernel gives

$$(E5) \quad v_2(x) = 6, \quad v_2(d) = 3, \quad u_d \equiv b_0 \pmod{8}, \quad u_x \equiv 1 - 2As^2b_0 \pmod{8}.$$

If s is even then $u_x \equiv 1$. If s is odd, then $A \equiv 5 \pmod{8}$ and $u_x \equiv 1 - 2b_0 \pmod{8}$. In the standard 2-adic Hilbert formula the exponent is

$$\epsilon(u_x)\epsilon(u_d) + 6\omega(u_d) + 3\omega(u_x) \pmod{2}, \quad \epsilon(u) = \frac{u-1}{2}, \quad \omega(u) = \frac{u^2-1}{8}.$$

For $b_0 \equiv 1, 3, 5, 7 \pmod{8}$, the corresponding u_x are 7, 3, 7, 3, and the exponent is respectively 0, 0, 0, 0. Thus the 2-adic symbol is always +1.

For every other odd $p \in S$, (E2) makes $2fq_1$ a square unit. Because α is a rational square, $d = 2\alpha fq_1$ is a square in $\mathbb{Q}_p$ even when $p \mid a$; hence $(x, d)_p = +1$. At infinity $d > 0$, so the real symbol is +1.

At the moving prime the kernel identity is

$$(x, d)_{q_1} = \left(\frac{A}{q_1} \right).$$

Every prime divisor of A is $1 \pmod{4}$, because $(2a)^2 \equiv -1$ modulo that prime. Reciprocity and (E2) therefore give

$$(E6) \quad \left(\frac{A}{q_1} \right) = \prod_{p^e \parallel A} \left(\frac{q_1}{p} \right)^e = \left(\frac{2f}{A} \right) = \left(\frac{2}{A} \right) \left(\frac{f}{A} \right) = (-1)(-1) = +1.$$

Here $A \equiv 5 \pmod{8}$ and $\left(\frac{f}{A} \right) = \left(\frac{A}{f} \right) = -1$. This is the compatibility identity that prevents a hidden collision between the frozen and moving places.

Finally define

$$k_p = \texttt{_110_exponent}(P, b_0, p), \quad N = \text{lcm}(8, \{p^{k_p} : p \in S\}, 4A).$$

The Taylor-exponent lemma freezes the S -symbols throughout $q_1 + N\mathbb{Z}$, while $4A \mid N$ freezes (E6). Since (E2) makes q_1 a unit at every prime dividing N , $\gcd(q_1, N) = 1$. This completes the class construction. Emergent places outside S have deliberately not been controlled; finding a member without a negative emergent symbol is the separate member-existence problem. $\square$

Remark 5.2 (The canonical parameter reproduces the 5-wall). For $a = 1$ one has $A = 5$ and $\tau^\dagger = 3/5$. The independently computed frozen and moving symbols reduce to

$$\text{Hilb}_5 = - \left(\frac{f}{5} \right) \left(\frac{q_1}{5} \right), \quad \text{Hilb}_{q_1} = \left(\frac{5}{q_1} \right) = \left(\frac{q_1}{5} \right).$$

They can both be +1 exactly when $\left(\frac{f}{5} \right) = -1$. For $f = w$ and the prime- b route unavailable, this fails exactly at $w \equiv 11, 19 \pmod{20}$, the independently derived canonical 5-wall of Theorem 9.3. The separate replay recovered all 60 walled grid cells and repaired all 60 after allowing a to vary. Thus the collision at $a = 1$ is exact, while class nonexistence is not.

6. THE VERIFIED QUANTITATIVE LAYER

The class construction reduces the residual local problem to a moving arithmetic progression. For an aligned class write

$$Q(k) = q_1 + kN, \quad b(k) = \varepsilon f Q(k),$$

with the controlled Hilbert symbols already fixed to be positive. A member is called *emergent-free* when the remaining odd-valuation places contribute no negative symbol. The purpose of the

quantitative layer is twofold: to certify this condition on the finite grid and to separate the exact local mechanism from the statistical model suggested by the resulting population.

6.1. L14: instance verification.

Proposition 6.1 (Finite instance of hypothesis H). ***PROVED (per row).** All 293/293 aligned classes in the finite grid have a certified emergent-free member; these comprise 190 L11 classes and 103 escape classes. Every stored row was replayed through the exact cofactor ladder and returned verdict zero.*

The proposition is deliberately finite. Its acceptance condition is an aligned class together with an exact zero verdict from the member ladder. A tied-conic or ramification computation may also be attempted as an auxiliary cross-check, but a budget refusal there is accepted and logged: it neither invalidates a ladder certificate nor supplies positive evidence. In particular, no refusal is promoted to a primality or solubility assertion. This distinction is essential because the general, all-cell form of hypothesis H remains open even though the displayed grid instance is complete.

6.2. L15: the exact remainder and counting laws. For a certified member, let R be the remainder of $P(b)$ after the frozen support, the fixed shell, and the smooth-emergent primes have been stripped. An independent replay gives the following exhaustive shape table.

Ladder shape	Rows	Exact form of R	Size data
proved-prime	197	one prime, never pC^2	median 48 digits; maximum 119
factorint	96	squarefree product, $e \in \{2, 3, 4\}$	multiplicities 76/15/5

TABLE 1. **PROVED (finite audit).** Shapes of the stripped remainder on all 293 closure rows.

Thus R is squarefree on every audited row and $R = 1$ never occurs. The prime rung is not a probable-prime shortcut: its singleton cofactor is accepted only after a proof of primality. On a factorint rung the integer e counts distinct emergent factors, and the exact factorization allows every corresponding Hilbert symbol to be checked. The two shapes therefore reflect two proof routes through the same local criterion, not two definitions of closure.

There is also an exact finite counting statement. The first certified member occurs at $k = 0$ in 207/293 classes, or 70.65%. This is **PROVED** as a statement about deterministic playback of the stored progressions. By contrast, interpreting the searches as samples from a stable member process is **EVIDENCE**: the measured per-prime-member emergent-free rate is approximately 0.040, and the observed waiting time is approximately 15.7 times the pure Bateman–Horn [1] first-prime prediction for the same progressions. The exact histogram may therefore be used without qualification inside the finite audit, while the rate and the comparison are empirical laws, not uniform density theorems.

6.3. L16: why the small layer has sieve shape. Fix an emergent odd prime p . On an aligned class, $b(k)$ is affine in k , and the sign at p is

$$(x_0, d_0)_p = \left(\frac{2\alpha b(k)}{p} \right).$$

On every checked recurring-prime row, this sign is determined by $k \bmod p$. The obstruction at p is consequently not the whole root set of $P(b(k))$ modulo p : it is the union of those root classes having the bad signature, further restricted to odd valuation,

$$\mathcal{B}_p = \{k : v_p(P(b(k))) \text{ is odd}\} \cap \{k \bmod p : \left(\frac{2\alpha b(k)}{p} \right) = -1\}.$$

For a simple bad root, Hensel lifting gives an exact Haar mass

$$\sum_{j \geq 0} \frac{p-1}{p^{2j+2}} = \frac{1}{p+1}.$$

This is the valuation-parity correction to a naive exclusion sieve: one must retain the odd levels of the Hensel tower rather than merely delete a residue class modulo p . A singular root has no such one-line factor. Its descendants are lifted recursively until the odd and even masses are certified with a residual bound; if that cannot be completed, the corresponding local product is explicitly labeled as an exclusion or partial result. Accordingly, the residue and valuation checks are **PROVED per row**; the claim that this small layer supplies the useful global layering model is **EVIDENCE**. No independence theorem is hidden in the product notation.

6.4. L17: matched validation. The full-grid residue test strengthens the preliminary structural probe. Across all 293 classes and the window $k \in [0, 119]$, it found zero residue-determination counterexamples on the identical 8,760-member sample used for evaluation. This is **PROVED** as a finite statement. The predictive use of the local masses is **EVIDENCE**; it was assessed only after the predictions and observations had been put on exactly the same member support.

Matched statistic	Model	Observed	Diagnostic
marginal odd hits	4481.3	4478	ratio 0.9993
hit-conditioned union	3477.9	3475	residual -2.9
permutation null	—	3475	$z = -0.09$; tails 0.61/0.53

TABLE 2. **EVIDENCE**. Matched small-layer evaluation on the same 8,760 members used on both sides.

The marginal statistic counts odd bad-root hits, so a member can contribute more than once; the union statistic asks whether a member is hit at least once. Their simultaneous agreement, together with the marginal-preserving permutation null, shows no detectable overlap or dependence discrepancy on this window. It does not prove independence outside the evaluated support.

A superficially different class-weighted comparison has estimand about $+0.0195$, at 3.63 standard deviations. That quantity compares Haar-uniform local masses in $\mathbb{Z}_p$ with the finite $k \in [0, 119]$ window, rather than comparing predictions and observations member by member on matched support. It is therefore identified as the Haar-window gap, not as a failure of the hit-conditioned model.

6.5. L17 cofactor layer: reciprocity and its limit. The large-cofactor layer has a narrower theorem than the small layer. **PROVED:** on the prime rungs, reciprocity forces the singleton cofactor sign to be $+1$ in all 197/197 cases; on the complete closure set, the cofactor-sign product is $+1$ on 293/293 rows. These are exactly the singleton and parity-product statements. The individual positive signs in factorint closure rows are selected by the closure condition, so their residue tables are descriptive data and cannot support an unconditional sign-distribution claim.

The attempted nonclosure measurement remains **INCONCLUSIVE**: only 11/400 sampled rows were resolved, and the predeclared power gate failed. The unresolved rows are refusals, not observations of either sign. Consequently the cofactor layer is proved at reciprocity level and no further; its nonclosure sign distribution remains measurement-limited. This is also why the matched small-layer success must not be silently upgraded into a complete rate theorem for emergent-free members.

7. THE CONDITIONAL RECORD AND THE SCHINZEL AUDIT

7.1. The statement, with its hypothesis exposed. The hypothesis used below is not an unnamed appeal to genericity. It is the following cellwise assertion, quoted in the form in which it is recorded in the ledger.

Definition 7.1 (CONDITIONAL: member hypothesis H). For every odd prime w and every rational z with $v_w(z) \geq 1$ there exists a verified aligned class for the cell (w, z) and a prime member $Q \equiv q \pmod{N}$ outside its finite excluded set such that the bad emergent set is empty. Explicitly, for every odd prime $\ell \notin S \cup \{Q\}$ with $v_\ell(P(\varepsilon f Q))$ odd, the local Hilbert symbol $(x_0, d_0)_\ell$ is $+1$. Here S is the frozen set of the class.

Theorem 7.2 (CONDITIONAL: Theorem C). *Assume H. Then $\mathbb{Q} \setminus \mathbb{Z}$ is Diophantine in $\mathbb{Q}$ with six unknowns, realized by the L6 formula*

$$F(z) := \exists b, s \left[(1 + 2s, b) \in \Phi_1^{\{2\}} \wedge \Theta_*(1 + 2s, b, h(1 + 2s, b, z^3), s) \right],$$

where, with $A = 1 + 4a^2$, $B = 2b$, and $\tau^\dagger = (1 + 2a^2)/A$,

$$\begin{aligned} \Theta_*(a, b, c, s) := \exists y, r \quad & [c^2 - Ay^2 - 16Br^2 = 16 - 16ABs^2] \\ & \vee [c^2 - Ay^2 - 16ABr^2 = 16A - 16A^2Bs^2] \\ & \vee [a^4(c^2 - Ay^2) + 4ABr^2 = 4A^2Bs^2 - 4A]. \end{aligned}$$

The three disjuncts are the cleared equations for $\tau = 0$, $\tau = 2a/A$, and $\tau^\dagger$, respectively, in the same witnesses (y, r) . Consequently $\mathbb{Z}$ is definable in $\mathbb{Q}$ by a formula with six universal quantifiers, $\mathbb{Z} = \mathbb{Q} \setminus (F \vee \mathfrak{m}_2)$, and $\text{efd}_{\mathbb{Q}}(\mathbb{Q} \setminus \mathbb{Z}) \leq 5$.

This is the ledger's exact conditional conclusion. It is *not* an unconditional definition and makes **no record claim**. Soundness of the expanded branch menu consumes no hypothesis; completeness is exactly H. Theorem 5.1 proves the class component uniformly, rather than on a finite horizon. The remaining member component follows from classical Schinzel H only after the chosen pair is proved admissible; uniform irreducibility of its octic is still open.

The quantifier saving is attributed precisely to Nicolas Daans, Philip Dittmann, and Arno Fehm, *Existential rank and essential dimension of diophantine sets*, arXiv:2102.06941v5, Theorem 1.4 (equivalently Corollary 5.11) [6]. Their intersection theorem turns the ranks three and two on the common base into $3 + 2 - 1 = 4$ witnesses; the two base coordinates then give six unknowns. The recorded rank–essential-dimension identity gives the bound five. These counts remain **CONDITIONAL** because the set equality to $\mathbb{Q} \setminus \mathbb{Z}$ uses H.

7.2. What is proved, and what Schinzel must supply. PROVED (uniform class side and linear side). Theorem 5.1 constructs an aligned class for every cell and freezes all controlled Hilbert symbols while restricting the moving prime to one reduced progression

$$q_1(t) = q_{1,0} + Nt, \quad \gcd(q_{1,0}, N) = 1.$$

The residue construction gives infinitely many proven-prime choices of the base $q_{1,0}$, and Dirichlet's theorem supplies infinitely many prime values of the progression. This is the linear, unconditional half of the member-existence problem; no probable-prime test or refusal is used as evidence.

OPEN/CONDITIONAL (cofactor side). Put

$$F_{\text{cell}}(t) = P(\varepsilon f q_1(t)),$$

using the primitive integer representative after removal of the fixed rational scale recorded in the audit. Requiring $F_{\text{cell}}(t)$ to be prime simultaneously with $q_1(t)$ is the two-polynomial instance of Schinzel's Hypothesis H [12] (or the corresponding Bunyakovsky-type prime-value input). The polynomial F_{cell} has degree eight. A prime cofactor is a sufficient route to an emergent-free member,

because the class has already frozen the controlled places and reciprocity forces the remaining single symbol. This sufficient prime-cofactor route must not be confused with a proof of H, whose conclusion asks only for one emergent-free member and is formally weaker. In the ledger’s shorthand, after using the progression prime itself as variable, the pair is written $\{t, F_{\text{cell}}(t)\}$.

7.3. PROVED finite audit of the Schinzel pair. The exact audit contains 293 canonical records, split into 190 L11 classes and 103 escape classes. The fields `F_irreducible`, `F_value_gcd`, and `pair_product_gcd` respectively certify irreducibility on 293/293 rows, value gcd one on 293/293 rows, and pair-product gcd one on 293/293 rows. Every irreducibility certificate is a degree-eight Frobenius certificate modulo a certified prime; both gcds were computed exactly on $0 \leq t \leq 2000$. The pair-product computation is already a certificate that no prime divides the product for every integer t : any such fixed divisor would divide the computed gcd. Thus all canonical pairs pass the fixed-divisor and irreducibility conditions. This is **PROVED** for the stored pairs, not a prime-value theorem.

The finite audit does not extrapolate to the classes just constructed for arbitrary rational cells. The L20 audit proves their normalization, coprimality, fixed-divisor, frozen-unit, symbol-constancy, and positive-lead conditions uniformly, but it leaves irreducibility of $P(a, z)$ open uniformly. Since affine substitution gives G irreducible over $\mathbb{Q}$ if and only if P is, this is the exact algebraic premise still missing before classical Schinzel H applies in every cell.

For a prime p dividing N , the corrected residue count is

$$\nu_p = \#\{t \in \mathbb{F}_p : q_1(t) \not\equiv 0 \pmod{p}, F_{\text{cell}}(t) \not\equiv 0 \pmod{p}\}.$$

The first condition is essential. Since $p \mid N$ and $\gcd(q_{1,0}, N) = 1$, the linear polynomial is the nonzero constant $q_{1,0}$ modulo p ; it must not be replaced by the parameter t . Consequently the count is p , rather than $p - 1$, whenever the cofactor has no root modulo p . The audit enumerated every $p \leq 1000$ dividing the row’s modulus and obtained exactly this full count in every occurrence.

primes p	records containing $p \mid N$	range of ν_p	mean density ν_p/p
2, 3, 5, 7	293 each	p	1
11, 37, 89, 97	13 each	p	1
13, 53	26 each	p	1
17	39	p	1
23	38	p	1
29, 41, 61, 73	28 each	p	1
43, 47, 67, 83	10 each	p	1

TABLE 3. Corrected admissible-residue counts for primes dividing the progression modulus.

These local counts neither assume independence nor estimate a density of simultaneous prime values. Their role is narrower and exact: together with the pair-product gcd, they rule out the elementary local obstruction for each stored Schinzel pair.

7.4. Residual certificates and the literature boundary. The six residual rows are deliberately segregated. They are rational or composite- b witnesses verified by exact branch reconstruction and tied status; they are not coerced into the prime family and are not evidence for Schinzel’s hypothesis.

OPEN (published boundary). The exact square-class reference is David Krumm, *Squarefree parts of polynomial values*, Journal de Théorie des Nombres de Bordeaux **28** (2016), no. 3, 699–724, arXiv:1407.4890 [10]. Its Theorem 1.3 and Propositions 3.4–3.5 give the unconditional degree-one and degree-two cases. Proposition 3.8 makes the degree-three case conditional on the Parity Conjecture for elliptic curves over $\mathbb{Q}$ —rank parity, not the sieve “parity problem.”

At prime arguments, the nearest unconditional results reach the cubic boundary without prescribing a square class: Helfgott proves square-free values of $f(p)$ for separable cubic f [8], while

cell (w, z)	(a, b)	τ	status
$(67, 2)$	$(1, 4757)$	$3/5$	PROVED pointwise
$(41, 1/7)$	$(25, 4985)$	$1251/2501$	PROVED pointwise
$(61, -2/7)$	$(25, 55571)$	$1251/2501$	PROVED pointwise
$(61, 2)$	$(25, -71431)$	$1251/2501$	PROVED pointwise
$(53, 1/3)$	$(-15, 9)$	$451/901$	PROVED pointwise
$(29, -2)$	$(-15, -29/5)$	$4055/2703$	PROVED pointwise

TABLE 4. The six residual witnesses: unconditional certificates for their individual cells only.

Reuss proves $(d - 1)$ -free values rather than prime or prescribed-square-class values [11]. Neither yields the required degree-eight conclusion.

The conic-bundle literature does not remove the gap. The unconditional Harpaz–Skorobogatov–Wittenberg theorem requires abelian constant fields in the bad fibres [7, Theorem 3.1 and Corollary 3.4]. The general Schinzel reference in *J. reine angew. Math.* **495** (1998), 1–28 is by Colliot-Thélène, Skorobogatov and Swinnerton-Dyer [3], not by Colliot-Thélène and Sansuc; the latter pair’s Schinzel paper is [2]. Thus no unconditional result is available for the degree-eight prime-value side or for the required degree-eight square-class side. The six-quantifier theorem above remains **CONDITIONAL**, and no unconditional record is claimed.

8. SCHINZEL’S HYPOTHESIS H: THE FIXED-CLASS IMPLICATION AND ITS GAP

The member hypothesis of Section 7 has two logically distinct parts. Theorem 5.1 now supplies an aligned class for every cell. For any fixed class satisfying the standard polynomial admissibility package, Schinzel’s Hypothesis H [12] supplies the member. The implication is proved below. What is not yet proved is that the constructed octic is irreducible for every cell, so the fixed-class theorem cannot yet be composed with the class theorem uniformly.

8.1. The operational contract. A place is recorded as an obstruction only when its valuation is odd *and* its Hilbert symbol is -1 : the kernel appends a place under exactly that conjunction, and the ladder returns *zero* when every remaining place is proved with symbol $+1$. Emergent-freeness therefore means *no place of symbol -1* , not *no odd valuation outside the frozen set*. This distinction is load-bearing: 197 of the 293 grid closures have exactly one odd-valuation outside prime whose symbol is forced $+1$ by reciprocity. The stronger reading would not follow from prime values, since a prime value deliberately creates one odd place.

8.2. The theorem. Fix a verified aligned class with data $(a, \varepsilon, f, q_1, N)$, put $Q(t) = q_1 + Nt$, $b(t) = \varepsilon f Q(t)$, and let $S = \{2, 3, 5, 7\} \cup \text{supp}(\alpha) \cup \text{supp}(\delta) \cup \text{supp}(f)$. Write the composed class polynomial as

$$F(t) = P(\varepsilon f(q_1 + Nt)) = cG(t),$$

with $c > 0$ and $G \in \mathbb{Z}[t]$ primitive with positive leading coefficient. The relevant hypothesis on c is that its *square class* is supported on S . Literal S -support is false in general: for $(w, z) = (3, 3/11)$, $a = 1$, $q_1 = 19$ one gets $c = 9072/15692141883605$, whose outside part is 11^{-12} . Only the square class matters, because an outside place of even valuation contributes Hilbert symbol $+1$.

Theorem 8.1 (Schinzel implies H per admissible class; PROVED implication). *Assume the class certificate (all symbols at S , at the moving prime Q , and at infinity equal $+1$ throughout the progression), assume the square class of c is supported on S and $G(t)$ is a p -adic unit for every $p \in S$, and assume Q and G are irreducible with positive leading coefficients whose product has no fixed prime divisor. Then Schinzel’s Hypothesis H for $\{Q, G\}$ implies that the class contains infinitely many members with ladder verdict zero.*

Proof sketch. Schinzel supplies infinitely many t with $Q(t)$ and $G(t)$ simultaneously prime; the side conditions (Cauchy bound, $b \neq 0, 1$, and finitely many forbidden coincidences) exclude only finitely many t . Put $R = G(t)$. The part of c outside S need not be trivial, but every one of its valuations is even. At the moving prime, $b \equiv 0 \pmod{Q}$ and the kernel identity gives $P(b) \equiv -\delta a^4 Z^4 A^2 \pmod{Q}$, a unit, so $R \neq Q$. Consequently R is the unique additional outside place of odd valuation; every other outside place has even valuation and $v_p(d) = 0$, hence symbol $+1$. The certificate pins the symbols at S , at Q , and at infinity to $+1$. Hilbert reciprocity forces $(x_0, d_0)_R = +1$. Thus no place carries symbol -1 , and the ladder verdict is *zero*. $\square$

8.3. Uniform admissibility status. For the class of Theorem 5.1, let $L(t) = q_1 + Nt$ and normalize $P(fL(t)) = cG(t)$ with $c > 0$ and $G \in \mathbb{Z}[t]$ primitive. The exact status is as follows.

condition	requirement	uniform status
(a)	irreducible polynomials with positive leading coefficients	positive lead PROVED; irreducibility of P (equivalently G) OPEN
(b)	$\gcd(q_1, N) = 1$	PROVED
(c)	$L(t)G(t)$ has no fixed prime divisor	PROVED
(d)	$G(t)$ is a p -adic unit at S and the frozen square classes are constant	PROVED

TABLE 5. Admissibility of the uniformly constructed square-branch classes.

For (a), the constant and leading coefficients of P are $P_0 = P_8 = 4a^8 AZ^4 > 0$, and the affine substitution $b = f(q_1 + Nt)$ preserves factorization over $\mathbb{Q}$. Thus G has positive lead and

$$G \text{ irreducible over } \mathbb{Q} \iff P \text{ irreducible over } \mathbb{Q}.$$

No uniform proof of the right-hand statement is known.

For (b), the prime support of N is exactly S , while the CRT base q_1 is a unit modulo every prime in S . For (c), primes in S never divide either value by the Taylor congruences. If $p \notin S$, the linear factor has one root modulo p ; a fixed divisor would force the degree-8 polynomial G to vanish at the other $p-1$ residues, hence $p \leq 9$. Every such prime already lies in S , a contradiction. For (d), the Taylor-exponent construction gives coefficientwise

$$G(t) \equiv G(0) \not\equiv 0 \pmod{p} \quad (p \in S \text{ odd}), \quad G(t) \equiv G(0) \not\equiv 0 \pmod{8}$$

and the analogous congruences for $L(t)$. The resulting ratios are local squares, so every frozen Hilbert symbol is constant for all integer t .

The modular irreducibility engine nevertheless gives exact finite certificates: 353/353 independently reconstructed grid classes and 706/706 legacy cell-branch rows have irreducible degree-8 reductions at certified primes. The same audit found zero mismatches in the uniform proofs above. These are per-row theorems, not a uniform irreducibility theorem.

A separate 24-row cross-check re-ran the proven-primality kernel end to end: every sampled member had one proved-prime ladder residual with symbol $+1$ and verdict *zero*.

Remark 8.2 (Exact scope). Schinzel supplies members only after a chosen class satisfies every row of Table 5. Uniform class existence and conditions (b)–(d), together with positivity in (a), are proved. Irreducibility in (a) is not. Therefore the present result is the implication

$$\begin{array}{c} \text{uniform irreducibility of the constructed } P \\ + \text{ Schinzel H} \end{array} \implies \begin{array}{c} \text{the six-quantifier} \\ \text{conditional conclusion.} \end{array}$$

not “Schinzel H implies the record.” Uniform irreducibility is the single residual algebraic gap.

9. THE STEP-(II) DENSITY LAW AND THE OFF-GRID FRONTIER

This section reports the two newest components of the program: an unconditional local bound on the sieve masses together with the exact decay law it implies, and the current state of the off-grid horizon.

9.1. An unconditional local bound. Fix an aligned class with parameters (ε, f, q_1, N) and write $F(k) = P(\varepsilon f(q_1 + kN))$ for its degree-8 class polynomial, and $u(k) = 2\alpha\varepsilon f(q_1 + kN)$ for the unit whose Legendre symbol decides bad signature. For an odd prime p let m_p denote the Haar mass in $\mathbb{Z}_p$ of the set of k with $v_p(F(k))$ odd and $\left(\frac{u(k)}{p}\right) = -1$: the local mass of the emergent obstruction.

Theorem 9.1 (Local sieve bound; PROVED). *For every aligned class of the grid,*

$$m_3 = m_5 = m_7 = 0, \quad \text{and} \quad m_p \leq \frac{8}{p+1} < 1 \quad \text{for all } p \geq 11,$$

the bound holding for singular roots as well, counted with multiplicity.

The vanishing at $p \in \{3, 5, 7\}$ is exactly the frozen-alignment condition of the class certificate: those places are controlled, and their symbols are pinned to +1 by construction. For $p \geq 11$ the bound is the root count of a degree-8 polynomial: at most eight residues can be roots, each simple root contributes Haar mass $1/(p+1)$ for odd valuation, and a singular root contributes at most its multiplicity times that mass, which is again bounded by the total degree. The bound was verified independently on every stored local pair: 102,439 pairs with $p \leq 10^4$ across the 293 classes, zero violations.

Theorem 9.1 settles a natural hope in the negative, and this is the important structural point.

Corollary 9.2 (No uniform positive mass; PROVED for the current classes). *For each of the 293 classes the splitting condition governing the bad-signature event has Chebotarev exponent $c = 1/2$: the class polynomial is irreducible with a replayed degree-8 certificate, and a replayed witness shows the bad unit is a nonsquare at a simple bad root. Consequently the truncated clean product obeys*

$$\prod_{p \leq X} (1 - m_p) \asymp (\log X)^{-1/2},$$

so no cutoff-uniform positive lower bound on the clean mass exists: the small-layer clean probability decays, slowly, to zero.

The measured exponent agrees: the root-harmonic estimate gives $c \approx 0.4778$, the product estimate $c \approx 0.4801$, and the same-member sieve between 10^4 and 10^5 gives $c \approx 0.4590$ (EVIDENCE).

The consequence for hypothesis H is a sharpening, not a defeat. The number of admissible prime members up to height X in a class grows like $X/(\log X)^2$ by the Dirichlet–Bateman–Horn heuristic [1], while each carries clean probability $\asymp (\log X)^{-1/2}$; the expected count of emergent-free members therefore still diverges, at rate $X/(\log X)^{5/2}$. What Corollary 9.2 rules out is the clean shortcut — a uniform positive constant — and what it leaves is a divergence statement of exactly the shape the measured counting law already exhibits (first emergent-free member at $k = 0$ in 207 of 293 classes).

9.2. The off-grid horizon. The grid is w prime, $w < 100$. Off-grid closures now exist for

$$w \in \{101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179\}, \quad u = -1,$$

together with $[131, [5, 1]]$, each certified by an aligned class and an exact ladder verdict *zero*. The routes are the L11 family, the canonical fixed-factor escape, and the non-canonical $a = 7$, $\tau = 0$ family. The last searched open cell, $[179, [-1, 1]]$, is now closed, so every off-grid cell tried to date has a replayed member. This is a finite-horizon statement only.

Two negative or delimiting findings carry structural content.

Theorem 9.3 (The canonical 5-wall; PROVED). *On the canonical escape route ($z = -w$, $a = 1$, $\tau = 3/5$, $f = w$), write $e = v_5(D)$. Then $v_5(x_0) = -3 - 2e$ is odd and $v_5(d_0) = 0$, so*

$$\text{Hilb}_5 = -\left(\frac{w}{5}\right)\left(\frac{q_1}{5}\right), \quad \text{Hilb}_{q_1} = \left(\frac{5}{q_1}\right),$$

and both equal $+1$ for some admissible q_1 if and only if $\left(\frac{w}{5}\right) = -1$. Since the L11 route requires a root of $1 + 4r^2 \equiv 0 \pmod{w}$, i.e. $w \equiv 1 \pmod{4}$, the canonical protocol is obstructed exactly for

$$w \equiv 3 \pmod{4} \text{ and } \left(\frac{w}{5}\right) = +1 \iff w \equiv 11, 19 \pmod{20}.$$

The criterion was audited on all 53 primes $w \in [101, 400]$ with both character branches of q_1 covered: 26 cells have the L11 route, 13 the canonical escape route, 14 are obstructed, and there were zero identity or classification mismatches.

The wall is a statement about the canonical parameter, not about the cells. In fact it is always breakable on the class side.

Theorem 9.4 (Every canonical 5-wall is breakable; PROVED). *Let $w \equiv 11, 19 \pmod{20}$ be prime and set $z = -w$, $f = w$. There is an odd integer a such that the $\tau = 0$ fixed-factor family has an aligned class for infinitely many positive primes q_1 . This theorem concerns class existence; it does not assert an emergent-free member. If a is fixed to 7, the class-side break exists exactly when $\left(\frac{w}{197}\right) = -1$.*

Proof. Because $w \equiv 3 \pmod{4}$, the character sum $\sum_{r \bmod w} \left(\frac{1+4r^2}{w}\right) = -1$ has no zero terms. Exactly $(w+1)/2$ residues therefore give character -1 ; choose one for $a \bmod w$. By CRT also impose $a \equiv 0 \pmod{\text{rad}(w^3+2)}$ and choose an odd lift. For $A = 1 + 4a^2$, $Z = -w^3$, and $D = 1 - Z - a^2Z^2$, the identity

$$4D \equiv (Z - 2)^2 \pmod{p} \quad (p \mid A)$$

then gives $\gcd(A, D) = 1$ while preserving $\left(\frac{A}{w}\right) = -1$.

At every prime occurring to odd order in A , prescribe $\left(\frac{-2\varepsilon w q_1}{p}\right) = +1$; impose the exact 2-adic class $\varepsilon w q_1 \equiv a \pmod{4}$; and make $-2A\varepsilon w q_1$ a square at each of 3, 5, 7 not dividing A . These independent local conditions define a reduced CRT class, so Dirichlet gives infinitely many primes q_1 in it. After finitely many exclusions and a real-sign threshold, every frozen, moving, fixed-factor, and infinite symbol is $+1$, and the exponent lemma freezes the resulting progression. The factor-by-factor identity

$$\prod_{p \mid A} (x, d)_p \left(\frac{A}{q_1}\right) = \left(\frac{-2\varepsilon}{A}\right) \left(\frac{w}{A}\right)$$

shows that for odd a the break criterion is $\left(\frac{w}{A}\right) = -1$. Putting $a = 7$ gives $A = 197$ and the stated fixed-parameter criterion. $\square$

Proposition 9.5 (Finite member closures; PROVED per row). *The non-canonical $\tau = 0$ family closes the obstructed cells $w = 131, 139, 151$ at $a = 7$, $q_1 = Q = 41$, $k = 0$. It also closes $[179, [-1, 1]]$ at $a = 7$, $\varepsilon = -1$, $f = 179$, $q_1 = Q = 251$, $k = 0$, with prime rung, tied status exact, and empty ramified set.*

These member rows are stronger than the class theorem only at their four individual cells. Their fixed factor $f = w$ divides $\text{num}(z)$, so they pass through the controlled non-coprimality door excluded by the generalized coprime wall. The kernel verdict uses the same tied-status, ramification, alignment, and complete cofactor ladder as the canonical rows; refusals in the surrounding scans are not evidence.

Remark 9.6 (Composite w). Composite w is not admissible in the architecture: w indexes a place, and the residue-field construction has no meaning for composite modulus — the break occurs already at the field constructor. The downstream arithmetic does generalize once w is decomposed into genuine prime places; as an illustration the pseudo-cell $[21, [-1, 1]]$ closes with the escape route at the honest prime $f = 3$ (member $k = 0$, $Q = 11$, factorint rung, tied and ramified checks exact). This is a clarification of scope, not an extension of the horizon.

10. CONCLUSION AND OUTLOOK

The logical chain now has one sharply isolated break. Uniform class existence is a theorem: every cell admits a square-branch aligned class, with infinitely many allowable base primes. For a fixed class that satisfies the standard two-polynomial admissibility package, Schinzel’s Hypothesis H supplies infinitely many emergent-free members. The normalization, coprimality, fixed-divisor, frozen-unit, symbol-constancy, and positive-leading-coefficient parts of that package are all proved uniformly. What is not proved uniformly is that the constructed octic $P(a, z)$ is irreducible over $\mathbb{Q}$. Thus classical Schinzel H alone does not yet imply the six-quantifier definition in this paper.

Three conclusions should govern further work.

- (1) *The residual algebraic problem.* Affine substitution and primitive rescaling preserve factorization, so the normalized class polynomial G is irreducible exactly when P is. Modular reductions give exact irreducibility certificates for every finite row audited, but a bounded ledger cannot settle arbitrary rational cells. A uniform proof for $P(a, z)$, or a proof that the freely chosen parameter a can always avoid the reducible locus while retaining the required character and congruence conditions, would close the only remaining admissibility gap.
- (2) *The member problem after admissibility.* The local sieve still has no cutoff-uniform positive mass: its clean product decays like $(\log X)^{-1/2}$, while the expected clean-prime count diverges like $X/(\log X)^{5/2}$. These facts explain why bounded search works often without proving a member. Once irreducibility is supplied, the fixed-class theorem invokes the classical simultaneous prime-value conjecture honestly; before then, invoking Schinzel hides a missing hypothesis.
- (3) *Walls and finite horizons.* Fixing $a = 1$ reproduces exactly the canonical 5-wall, but varying a breaks every such wall on the class side. The fixed choice $a = 7$ works only on its own quadratic-character branch. The cell $[179, [-1, 1]]$ now has a replayed ladder-zero member, so all searched off-grid cells are closed; this remains a finite collection of instance theorems, not an all-cell member theorem.

The published record therefore remains the refereed ten-quantifier definition. The result here is a conditional architecture with a proved class side, a proved fixed-class Schinzel implication, and one explicit uniform irreducibility gap. No stronger conclusion is claimed.

REPRODUCIBILITY

All certificates in this paper are produced by a stdlib-only kernel with a proven-primality discipline: deterministic Miller–Rabin below 3.3×10^{24} , Pocklington with the Brillhart–Lehmer–Selfridge relaxation above, and explicit refusal labels that never count as evidence. The companion paper documents the engine, the artifact schemas, the exact reproduction commands, and the precise scope of what is regenerable from checked-in scripts.

REFERENCES

- [1] P. T. Bateman and R. A. Horn, *A heuristic asymptotic formula concerning the distribution of prime numbers*, Mathematics of Computation **16** (1962), 363–367.
- [2] J.-L. Colliot-Thélène and J.-J. Sansuc, *Sur le principe de Hasse et l’approximation faible, et sur une hypothèse de Schinzel*, Acta Arithmetica **41** (1982), no. 1, 33–53.

- [3] J.-L. Colliot-Thélène, A. N. Skorobogatov and Sir P. Swinnerton-Dyer, *Rational points and zero-cycles on fibred varieties: Schinzel's hypothesis and Salberger's device*, Journal für die reine und angewandte Mathematik **495** (1998), 1–28.
- [4] G. Cornelissen and K. Zahidi, *Topology of Diophantine sets: remarks on Mazur's conjectures*, in: Hilbert's Tenth Problem: Relations with Arithmetic and Algebraic Geometry (Ghent, 1999), Contemporary Mathematics **270**, American Mathematical Society, Providence RI, 2000, pp. 253–260; arXiv:math/0006140.
- [5] N. Daans, *Universally defining $\mathbb{Z}$ in $\mathbb{Q}$ with 10 quantifiers*, Journal of the London Mathematical Society (2) **109** (2024), no. 2, e12864, 20 pp.; doi:10.1112/jlms.12864; arXiv:2301.02107.
- [6] N. Daans, P. Dittmann and A. Fehm, *Existential rank and essential dimension of diophantine sets*, preprint, arXiv:2102.06941v5 (2021–2024).
- [7] Y. Harpaz, A. N. Skorobogatov and O. Wittenberg, *The Hardy–Littlewood conjecture and rational points*, Compositio Mathematica **150** (2014), no. 12, 2095–2111; doi:10.1112/S0010437X14007568; arXiv:1304.3333.
- [8] H. A. Helfgott, *Square-free values of $f(p)$, f cubic*, Acta Mathematica **213** (2014), no. 1, 107–135; doi:10.1007/s11511-014-0117-2; arXiv:1112.3820.
- [9] J. Koenigsmann, *Defining $\mathbb{Z}$ in $\mathbb{Q}$* , Annals of Mathematics (2) **183** (2016), no. 1, 73–93; doi:10.4007/annals.2016.183.1.2; arXiv:1011.3424.
- [10] D. Krumm, *Squarefree parts of polynomial values*, Journal de Théorie des Nombres de Bordeaux **28** (2016), no. 3, 699–724; doi:10.5802/jtnb.959; arXiv:1407.4890.
- [11] T. Reuss, *Power-free values of polynomials*, Bulletin of the London Mathematical Society **47** (2015), no. 2, 270–284; doi:10.1112/blms/bdv007; arXiv:1307.2802.
- [12] A. Schinzel and W. Sierpiński, *Sur certaines hypothèses concernant les nombres premiers*, Acta Arithmetica **4** (1958), 185–208; erratum, ibid. **5** (1958), 259.
- [13] Z.-W. Sun, *$\mathbb{Q} \setminus \mathbb{Z}$ is diophantine over $\mathbb{Q}$ with 7 unknowns*, preprint, arXiv:2607.28606 (unrefereed).